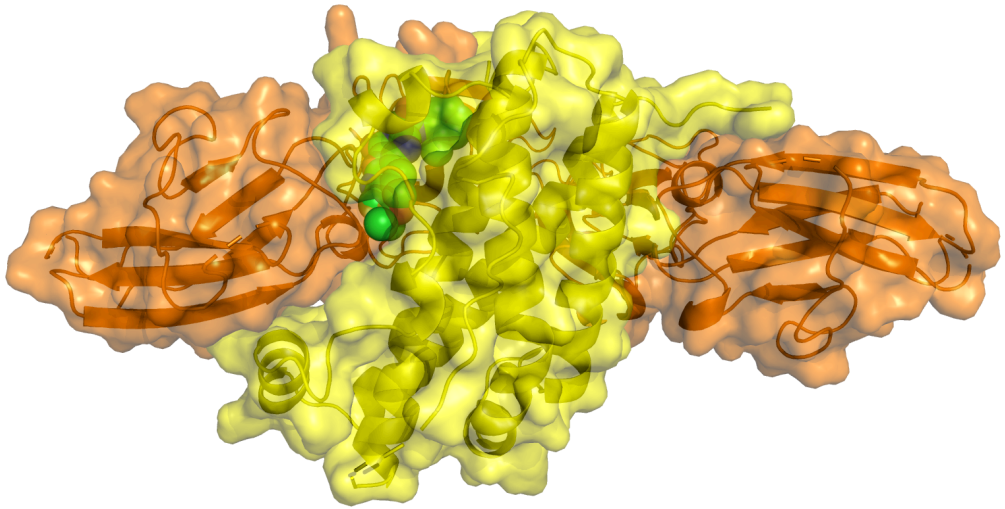
Building a Biotech Startup: the Synsight Story

International Business Administration

Cyril Bauvais, PhD



1.1 About me

From computational chemistry to company building

Cyril Bauvais, PhD

Co-founder and CEO of Synsight, 2013-2024

- PhD in chemoinformatics (Paris), postdocs in Chatenay-Malabry and Heidelberg
- IT production engineer for financial companies
- Research scientist in a Paris biotech (in silico toxicity prediction)
- Management training for innovative companies, HEC Paris

Université Paris Cité

UNIVERSITÄT HEIDELBERG ZUKUNFT SEIT 1386

HEC PARIS

Synsight company presentation 2023; Synsight deck 2022.

1.2 Today's case: one startup, many jobs

The areas we will cross in the Synsight story

Part 2

Industry

How pharma creates and buys innovation

Part 3

Creation

Founders, legal form, premises

Part 4

Strategy

Business model and its change

Part 5

IP

Patents, licences, data

Part 6

Financing

Grants, loans, equity

Part 7

International

Partners and investors in China

Part 8

Team

Recruiting, advisers

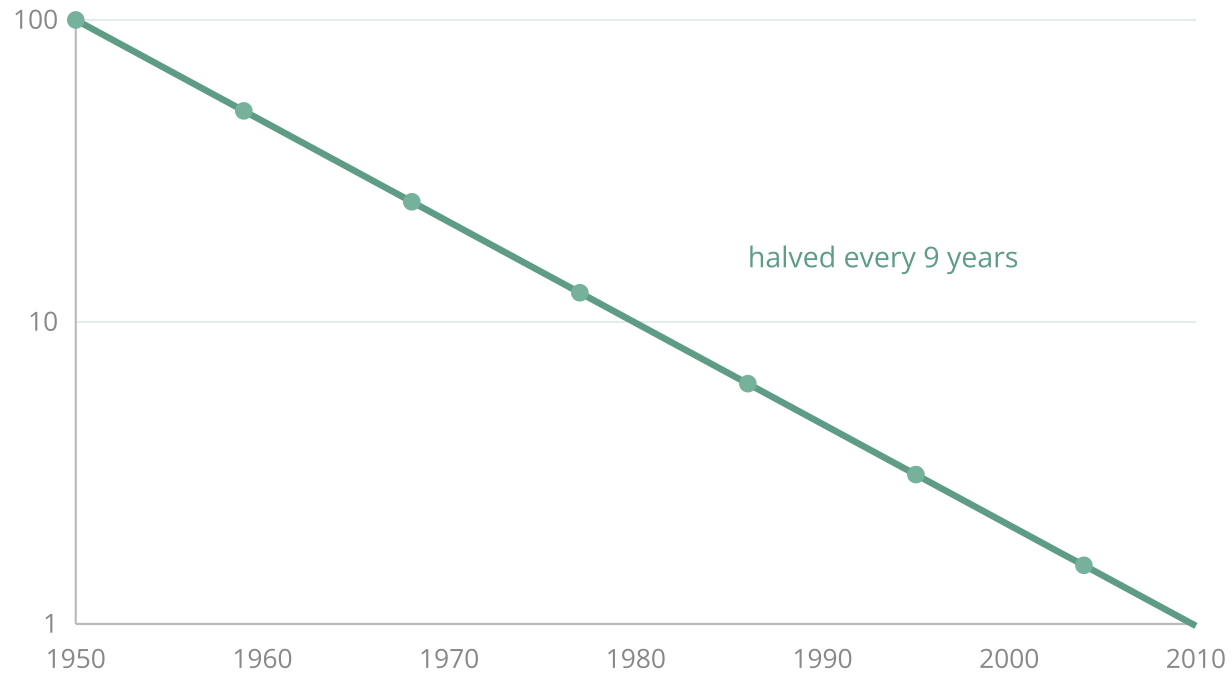
Part 9

Exit

Acquisition and integration

2.1 Why new medicines cost so much

R&D productivity has been falling for decades



New drugs approved per $1B of R&D, index 1950 = 100, log scale. Illustration of the trend measured by Scannell et al., 2012.

$985M

Median R&D investment to bring one new drug to market

FDA approvals 2009-2018, including failures

7.9 %

Drug candidates entering phase 1 that reach approval

2011-2020

- Easier targets have already been addressed, and each approved drug also pays for the projects that failed.

Wouters et al., JAMA 2020; BIO, Informa, QLS, Clinical Development Success Rates 2011-2020; Scannell et al., Nat Rev Drug Discov 2012.

2.2 The patent cliff

Big pharma must replace revenue that is about to disappear

- A patent lasts 20 years from filing; development uses a large part of it.
- At expiry, generics or biosimilars take most of the sales.
- **Over $230B** of US sales exposed between 2025 and 2030.
- Pharma companies license or buy projects from biotechs to refill pipelines.

WTO TRIPS Agreement, art. 33; GlobalData, 4 July 2025.

2.3 From target to patient

The stages of drug development

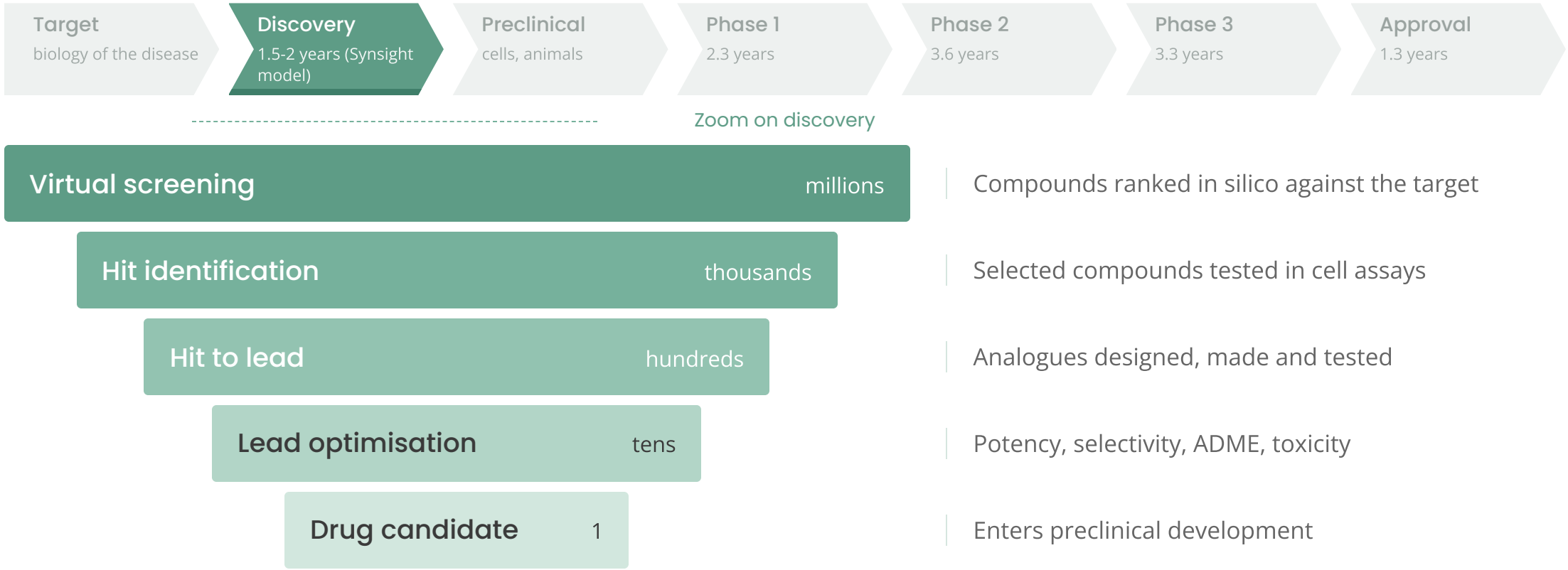


- From discovery to approval: 10 to 15 years, and $1 billion to $2.6 billion per approved drug, failures included.
- Only 7.9 % of candidates entering phase 1 are approved; phase 2 is the largest hurdle.

Durations: BIO, Informa, QLS 2011-2020; Synsight model 2024. Costs per trial: ASPE (US HHS) 2014. Totals: PhRMA 2020; JAMA 2020.

2.4 The drug discovery funnel

From a virtual library to a single drug candidate



- At Synsight, AI selected compounds so that only **10 to 20 %** of a chemical library had to be tested in the lab.

Orders of magnitude for illustration. Synsight figure: company deck 2023, p.12.

2.5 Two business models for a discovery startup

Own pipeline or partnered discovery

Own pipeline

- The startup discovers, patents and develops its own molecule
- Value is realised at licensing, often around phase 1
- Revenue: licence fee, milestones, royalties
- Requires large equity funding over several years

Partnered discovery

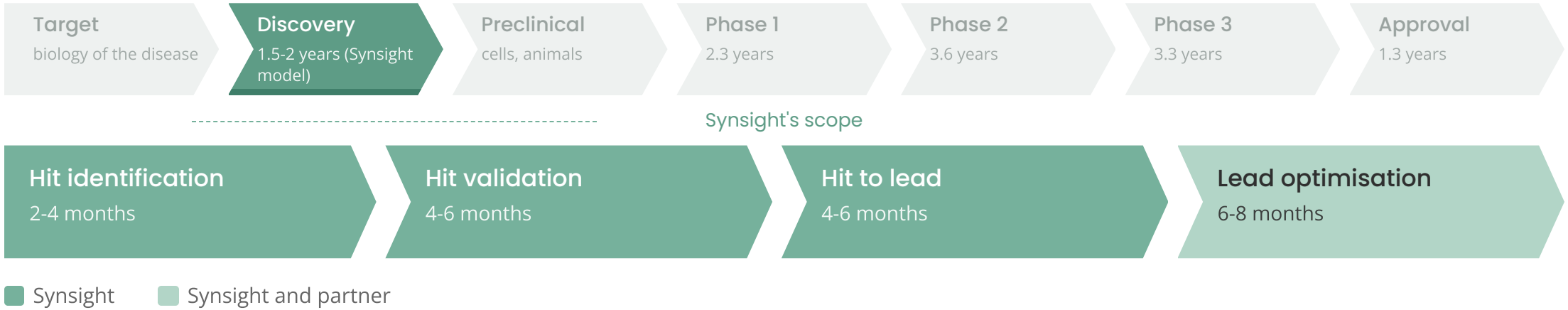
- The startup finds molecules for a pharma partner's target
- Upfront payment at signature
- Milestone payments on success: preclinical, first clinical data
- Earlier revenue, upside shared with the partner

- Example: Lilly and Anima Biotech, 2018: **$30M** upfront, $14M research funding, up to **$1.05B** in milestones.

BioSpace, 23 July 2018.

2.6 Synsight's place in the value chain

Drug discovery collaboration model, 2024



- Synsight: AI and computer-aided design, MTBench and cell imaging assays.
- Partner or subcontractors: chemical synthesis, ADME, in vivo studies.
- Each step closes on scientific criteria: potency, specificity, no cytotoxicity.

Synsight company deck, Feb 2024, p.20.

3.1 Two founders, 2013

Complementary profiles, explicit roles



Cyril Bauvais, CEO

PhD in chemoinformatics

- Strategy and business development
- Finance
- Relations with non-scientific stakeholders

Université Paris Cité



Guillaume Bollot, CSO

PhD in computational chemistry

- Scientific strategy
- Team coordination and HR
- IT and lab platform

UNIVERSITÉ DE GENÈVE

UCLA

- Both had worked in a French biotech before creating Synsight in December 2013.

Synsight company presentation 2023, p.3-4.

3.2 Setting up the company

Legal form, premises, ecosystem

- A simplified joint-stock company (SAS), the usual form for French startups.
- Hosted by the University of Evry in an Inserm unit: 63 m2 of offices, lab access.
- Renewable 3-year agreement, access to a high-content microscope.
- Part of the Genopole biocluster in Evry.



University of Evry. Photo: Cyrilb1881, CC BY-SA 3.0, Wikimedia Commons.

Synsight company presentation 2023, p.4.

4.1 2013-2018: a consulting company

Computer-aided drug design services

> 800 k€

Cumulative revenue 2013-2018

Consulting in computer-aided drug design

2

Founders delivering the work

No external funding needed

3

Client types: pharma, biotech, public research

Sanofi, Servier, L'Oreal, French cancer research centres

- Profitable from the start, but revenue grew only with the time sold.
- No asset was built: the results belonged to the clients.

Synsight company presentation 2023, p.2; BIO-Europe pitch 2022.

4.2 2018: changing the business model

From selling time to building assets

Before: services (2013-2018)

- Revenue from day one
- Growth limited by the size of the team
- No intellectual property owned
- Financed by its own revenue

After: technology and own programmes

- Platform combining AI and cell biology
- Patents, data and drug programmes as assets
- Longer cycle before revenue
- Needs grants and investors

- Trigger: an exclusive licence on a cell-based technology, MTBench, and the addition of biology to the platform.

Synsight company presentation 2023, p.2.

4.3 Anatomy of a collaboration proposal

Draft term sheet with a Chinese biotech partner, November 2022



Paid per task If the proof of concept succeeds

- Each task paid 50 % at start and 50 % on delivery.
- The partner could own the molecules; Synsight kept its technology.
- Six months of priority for the partner to negotiate a licence.

Synsight draft term sheet, research services agreement, 7 Nov 2022 (partner not named).

4.4 Business cases

Pharma, biotech and academic partners, 2013-2024

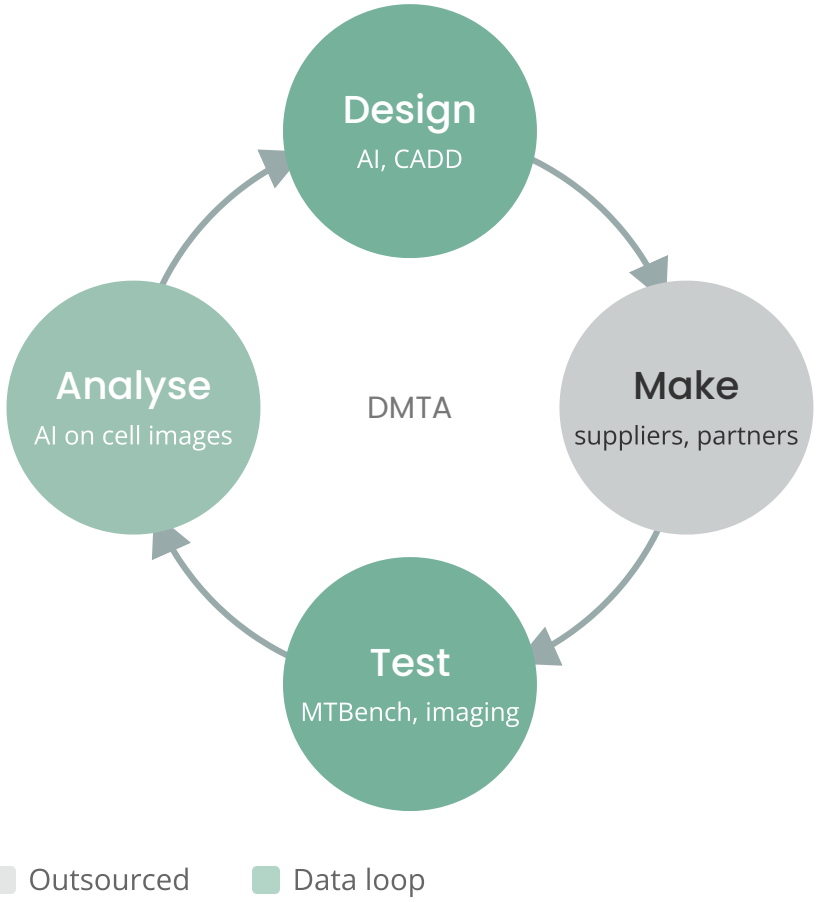


Synsight deck, Feb 2024, p.5 and p.18. Clients, collaborations and partnership discussions.

5.1 The design, make, test, analyse cycle

How drug discovery iterates

- **Design**: propose molecules, on a computer or with AI.
- **Make**: synthesise them, by chemists or robots.
- **Test**: measure their effect, on proteins or cells.
- **Analyse**: learn from the results and design the next round.



Solution overview, the dry lab and wet lab cycle (deck_Synsight_LR, p.11)

1.5.1 Solution

Overview -> Technologic cycle … combining AI and cell imaging for drug discovery

Training of AI models + Computer Aided Drug Design

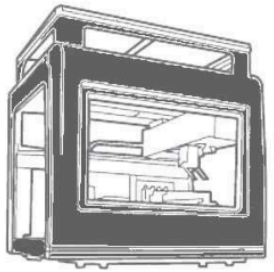
*Data from computer vision analysis are used to train AI models. These AI models **can then design better molecules***





Library of molecules

***Molecules** are ordered/synthesised and sent to our wetlab*





*Molecules are **mixed** with MTbench cells and mixed with Cell painting cells*

Recommended molecules

WETLAB

DRYLAB

Selected molecules & DATA

Automated data transfert



*Results of mixing is **hosted** by a robot, waiting to be imaged*

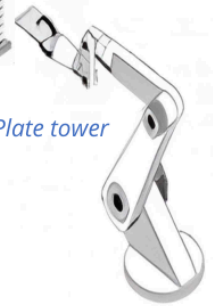
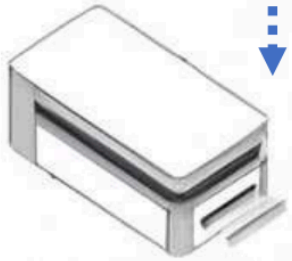


Plate tower



*Molecules are **tested experimentally** on cell painting and MTbench with **High-Content Imager***

Automated data transfert



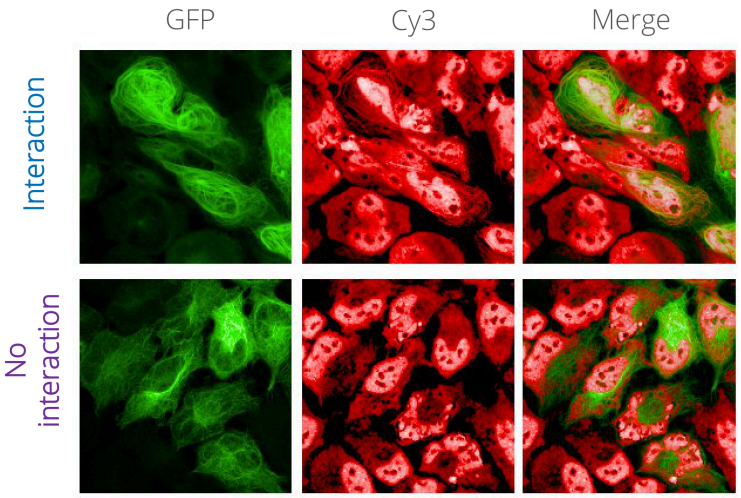
*Images of cells are analyzed to build **models of correlation** between molecules and their effect on the cells*

Automated computer vision analysis

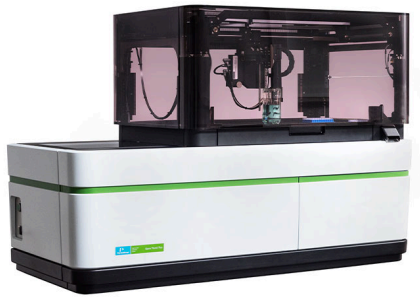
5.2 MTBench: measuring interactions inside living cells

A patented cell-based assay

- A bait protein is anchored on the cell's microtubules.
- Its partner, protein or RNA, co-localises on the microtubules when they interact.
- A compound that disrupts the interaction removes the signal, measured over thousands of cells.



Top: interaction. Bottom: no interaction. HeLa cells, Synsight high-content imager.



Opera Phenix Plus high-content imager. Manufacturer photo (PerkinElmer).

5.3 Intellectual property as an asset

Licence, patents, data

- The MTBench patent belonged to public research (Inserm, University of Evry).
- Synsight obtained an exclusive, worldwide licence on the patent and the know-how.
- By 2023: 2 patent families on MTBench and its improvement, and a proprietary database of 90 million compounds.

Synsight company presentation 2023, p.4; Synsight deck 2023, p.2.

6.1 Financing a deeptech startup

Synsight combined five sources of funding

Revenue

Customers

Consulting, then paid collaborations

Bpifrance

Loans and grants

Innovation loan 2019, innovation project 2020, Deep Tech aid 2022

Tax

Research tax credit

CIR on R&D expenses, pre-financed

Public research

Research grants

ANR project SPIRIT, CIFRE PhD

Equity

Investors

300 k€, then 803 k€

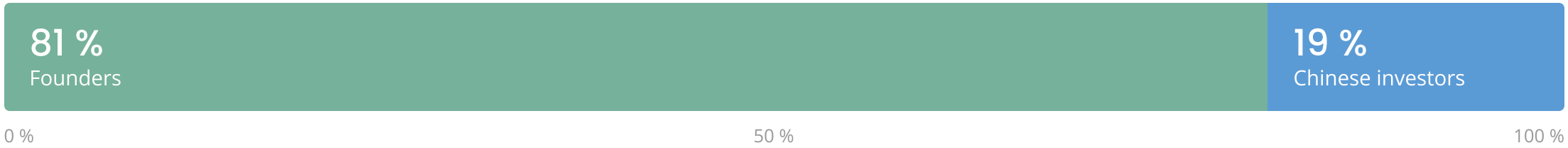
- Non-dilutive money first: it does not reduce the founders' shares.

Synsight company presentation 2023; Bpifrance contracts 2019-2022; ANR and ANRT agreements.

6.2 Raising equity

Two rounds with Chinese investors

Shareholding in 2023



- First round: 300 k€ (2018-2020), to set up the platform and the first programmes.
- Second round: 803 k€ (December 2021), to develop the platform and the programmes.
- After two rounds, the founders still held 81 % of the company.

Synsight company presentation 2023, p.2; Synsight deck 2023, p.2.

6.3 Raising money in China

The process as we practised it



- In China, the 'BP' is the pitch deck itself, not a long business plan.
- Rule of thumb heard at the time: valuation = amount raised / 10 to 15 %.

Internal note on fundraising in China, Synsight, May 2022.

6.4 What investors ask for

Feedback from Bpifrance's investment team, February 2023

1

Patients

Which patients and which diseases are targeted

2

Revenue model

Upfront, royalties, services: pricing and split over 4-5 years

3

Roadmap

Milestones on the critical path to a clear target

4

Amount

Exact amount, consistent with the business plan

5

Use of funds

Split between capex, opex, R&D, business development

6

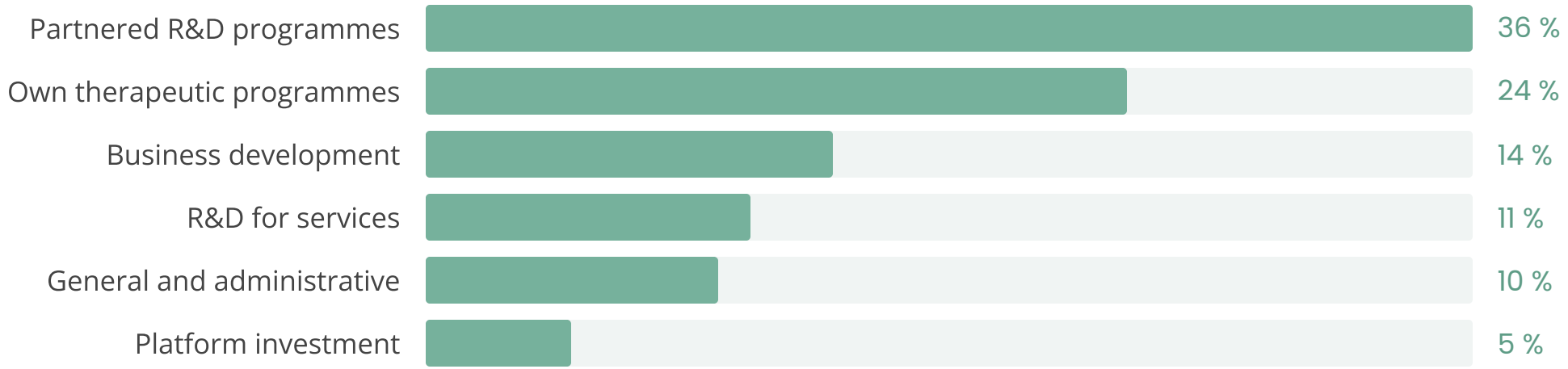
Financials

5-year P&L, financing plan, monthly cash plan

Bpifrance dealflow feedback to Synsight, 16 Feb 2023.

6.5 The 5 M€ plan

Planned use of proceeds of a Series A, 2022-2024



- 60 % of the money planned for R&D programmes, a typical profile for a discovery biotech.

Synsight deck 2022 (EU, non-confidential), p.25.

7.1 Going international: China

Partners, investors, a project of local company



Chongqing airport, December 2023: 'Western China Science City'.



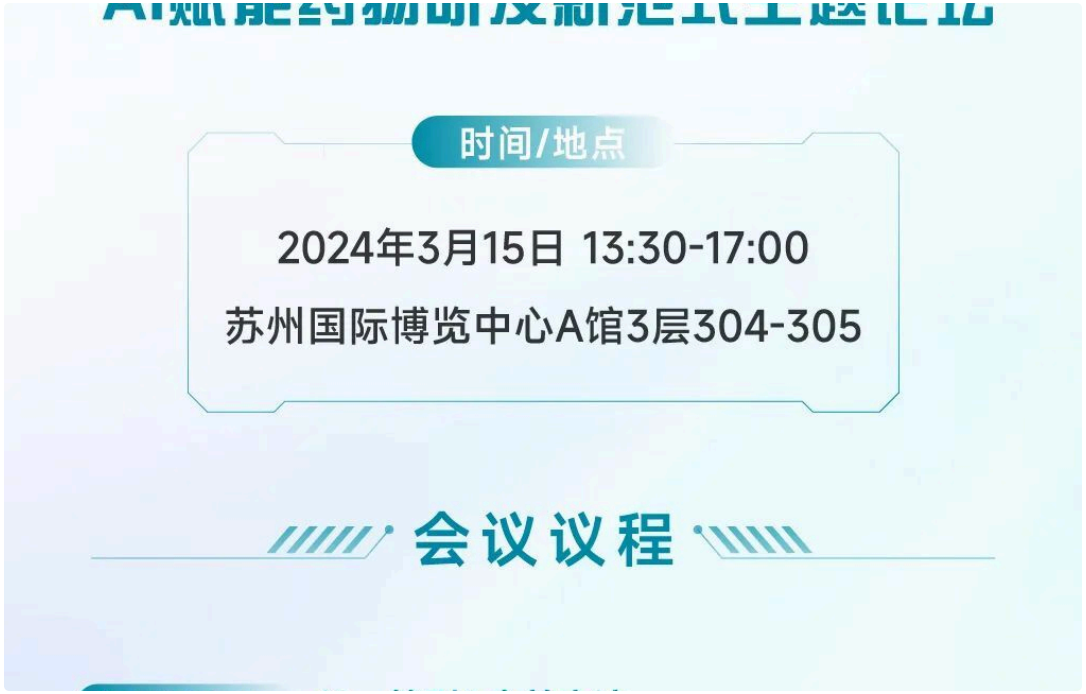
Innovation and entrepreneurship contest, Anhui province, Hefei, 2 December 2023.

- A Chinese investor in the capital and a partnership manager dedicated to China.
- Several trips: Chongqing, Hefei, Nanjing, Suzhou; a project to set up a company in China, not carried out.

Photos: Synsight, 2023.

7.2 Suzhou, March 2024

BIOCHINA forum on AI and drug discovery



Programme of the AI and drug discovery forum, BIOCHINA, Suzhou, 15 March 2024.



Visit of an innovation centre, Hefei, December 2023.

- Synsight presented a Chinese version of its pitch deck at BIOCHINA.
- Applications to local programmes, such as the Suzhou leadership programme for innovation and entrepreneurship.

Photos and documents: Synsight, 2023-2024.

8.1 Advisory board and experts

Borrowing credibility and know-how

- The two co-inventors of MTBench: an Inserm unit director and the University of Evry president.
- A computer vision expert (ENS) and a medicinal chemist with 32 years in pharma.
- Senior physicians in oncology and haematology.
- Advisers in finance and business development.

Synsight deck 2022, p.6 and p.23; Synsight deck 2023, p.7 and p.24.

TEAM



Cyril Bauvais

Co-founder & CEO

PhD in Chemoinformatics

Management, business creation and funding expert

Biotech R&D in drug discovery



Guillaume Bollot

Co-founder & CSO

PhD in Computational Chemistry

20 years' experience in computational chemistry

Biotech management



Aixiao Li

Business developer

PhD in Theoretical Chemistry

Partnerships and new markets developer

Founder of DAOSHANG consulting company

Team of 9 employees with 30% female ratio.

Future hires: COO, CFO, CMO, medicinal chemistry expert.



Julien Duez

Senior biologist

PhD in biology

Experience in early drug discovery and development



Amandine Molliex

Research scientist

PhD in Cell & Molecular Biology

Experience in cell biology



Ethan Cohen

AI Research scientist

PhD candidate in AI

Experience in data science applied to drug discovery

THE CORE TEAM



Anais Max

Marketing & Communication

Experience in communication & marketing for biotech



Etienne Lallemand

Chemoinformatics engineer



Maxime Sanchez

AI Engineer

THE ADVISORY BOARD



David Pastré

PhD in Optical Physics

Director of the Research unit INSERM U1204

Co-inventor of MT-Bench technology

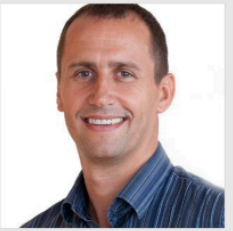


Patrick Curmi

MD & PhD Cell Biology

President of Université d'Evry – Paris-Saclay

co-inventor of MT-Bench technology



Auguste Genovesio

PhD in Computer Science

Director of the Research Unit "Computational Imaging & Bioinformatics at Ecole normale supérieure"



Denis Carniato

Senior Medicinal Chemist consultant from pharma industry

Medicinal Chemistry expert



Fabio Sterpone

PhD in Biophysics

Research director at CNRS, specialist in multi-scale modeling of biophysical processes. Biophysics expert



Ling Yue

China devel. plan strategy

Sponsor representative;

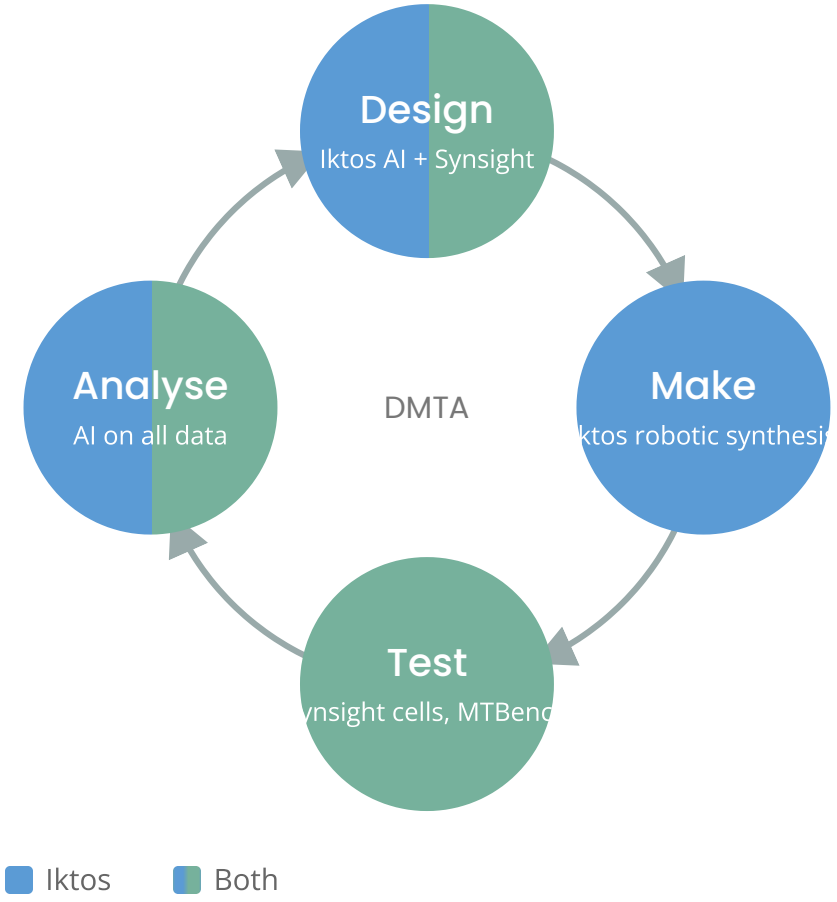
Chartered Financial Analyst

Close collaboration with Synsight

9.1 The acquisition by Iktos

Closing the design, make, test, analyse cycle

- Synsight did design and test; Iktos did design and make.
- Together they close the design, make, test, analyse cycle.
- Iktos: founded in 2016, generative AI for chemistry, about 70 people in 2024.

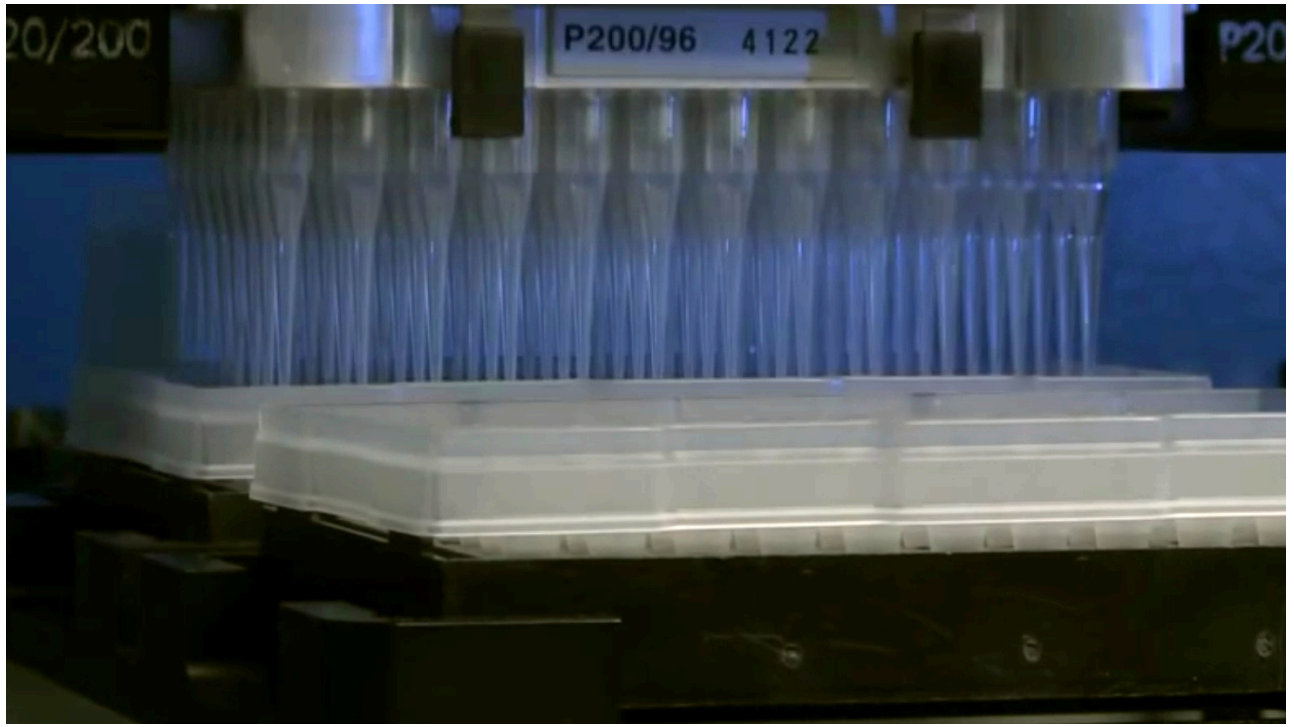


Iktos acquisition offer, June 2024.

9.2 After signing: the transition

In deeptech, the knowledge sits in people

- The CEO of Iktos asked the founders to stay for a few months.
- Goal: integrate the team, transfer know-how, align tools and processes.
- Keeping key people is part of the value of the deal.



Liquid handling robot. Photo: NIAID, public domain, Wikimedia Commons.

10.1 One startup, many jobs

What the Synsight story required

Industry

Market

Understand why pharma buys innovation

Strategy

Business model

Services, platform, own programmes

Legal

Contracts and IP

Licence, term sheets, agreements

Finance

Funding

Grants, tax credit, equity, cash plan

International

China

Investors, partners, local rules

People

Team

Recruiting, advisers, outsourcing

Sales

Partnerships

Pitch, negotiate, deliver

M&A

Exit

Fit, negotiation, integration

10.2 Conclusion

Synsight in five dates

2013
Creation, consulting in computer-aided drug design

2018
Exclusive MTBench licence, switch to a biotech model

2019–2022
Bpifrance, tax credit, ANR; equity rounds of 300 k€ and 803 k€

2022–2023
Collaborations, China, preparation of a 5 M€ round

2024
Acquisition by Iktos

- The business model determines how much money is needed, and from whom.
- In deeptech, the assets are intellectual property, data and people.
- An acquisition is driven by strategic fit: what the buyer cannot build quickly.